

SECTION 5.3: THE FUNDAMENTAL THEOREM OF CALCULUS (DAY 2)

1. The Fundamental Theorem of Calculus (part 1):

$$\frac{d}{dx} \left[\int_a^x f(t) dt \right] = f(x)$$

2. Find the derivative of each function below.

(a) $g(x) = \int_{-1}^x t^2 e^t dt$

$$g'(x) = x^2 e^x$$

(b) $h(x) = \int_0^{x^2+1} \sin(t) dt$

$$h'(x) = \sin(x^2+1) (2x)$$

3. Let $f(x) = 4x$. Find two different antiderivatives of $f(x)$. Call them $F_1(x)$ and $F_2(x)$.

$$F_1(x) = 2x^2, \quad F_2(x) = 2x^2 + 1$$

4. Let $G(x) = \int_1^x 4t dt$.

(a) What do $G(x)$, $F_1(x)$ and $F_2(x)$ all have in common?

All have $y = 4x$ as their derivative.

(b) Find $G(1)$, $F_1(1)$ and $F_2(1)$. ~~You will have to find $G(1)$ by geometry.~~

$$F_1(1) = 2 \cdot 1^2 = 2$$

$$G(1) = \int_1^1 (4t) dt = 0$$

$$F_2(1) = 2 \cdot 1^2 + 1 = 3$$

(c) Find $G(3)$, $F_1(3)$ and $F_2(3)$. You will have to find $G(3)$ by geometry.

$$G(3) = \int_1^3 (4t) dt = 2 \cdot 4 + 8 \cdot 2 \cdot \frac{1}{2} = 16$$

$$F_1(3) = 18 \quad F_2(3) = 19$$

(d) Using your answers above, find $G(3) - G(1)$ and explain what it means geometrically about the curve $y = 4x$.

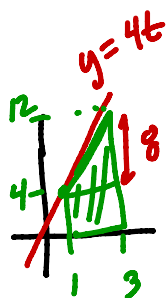
$$G(3) - G(1) = 16. \text{ Green shaded area.}$$

(e) Find $F_1(3) - F_1(1)$ and $F_2(3) - F_2(1)$.

$$F_1(3) - F_1(1) = 18 - 2 = 16 \quad F_2(3) - F_2(1) = 19 - 3 = 16$$

(f) What do parts (d) and (e) indicate about how you can calculate the (signed) area under a curve $f(x)$ on an interval $[a, b]$?

Pick any antiderivative of the integrand, $F(x)$.
 $F(b) - F(a)$ gives the area of $\int_a^b f(t) dt$
 net signed



5. The Fundamental Theorem of Calculus (part 2):

$$\int_a^b f(x) dx = F(b) - F(a) \text{ for } F'(x) = f(x) \\ \text{and } f(x) \text{ is continuous on } [a, b]$$

6. Evaluate the integrals.

$$(a) \int_0^{\pi/4} 5 \sec^2(x) dx \\ = 5 \tan(x) \Big|_0^{\pi/4} \\ = 5 \tan(\pi/4) - 5 \tan(0) = 5$$

$$(b) \int_{-1}^3 x + e^x dx = \left[\frac{1}{2} x^2 + e^x \right]_{-1}^3 \\ = \left(\frac{3^2}{2} + e^3 \right) - \left(\frac{(-1)^2}{2} + e^{-1} \right) = \frac{9}{2} + e^3 - \frac{1}{2} - e^{-1} \\ = 4 + e^3 + \frac{1}{e}$$

7. Find the average value of $f(x) = x^2$ over the interval $[0, 3]$.

$$\text{ave value} = \frac{1}{3-0} \int_0^3 x^2 dx = \frac{1}{3} \cdot \left[\frac{1}{3} x^3 \right]_0^3 = \frac{1}{9} \cdot 3^3 - \frac{1}{9} \cdot 0^3 = 3$$

8. Assume the velocity of an object thrown directly up into the air is given by $v(t) = 20 - 9.8t$ where v is measured in meters per second and t is measured in seconds.

(a) Evaluate $\int_0^1 v(t) dt$

$$\int_0^1 v(t) dt = \int_0^1 (20 - 9.8t) dt = \left[20t - 4.9t^2 \right]_0^1 = 20 - 4.9 = 15.1 \text{ m}$$

(b) Evaluate $\int v(t) dt$

$$\int v(t) dt = 20t - 4.9t^2 + C$$

(c) Explain why you do not have enough information to find the height of the object exactly?

We don't know the height of the object when it was thrown.

(d) Explain, in the context of the problems what part (a) and part (b) represent.

Ⓐ The change in height of the object in the first second. That is, the object is 15.1 m higher at 1 second than it was at 0 sec.

Ⓑ An equation for position (or height) but w/o more information, we can't find it exactly.